

Multi-View VLM Verification for Semantic Object Extraction from a Terrestrial Laser Scan Toward TOSM-Based Robot Environment Models

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Abstract—Semantic environment models such as the Triplet Ontological Semantic Model (TOSM) give indoor mobile robots the symbolic, explicit, and implicit knowledge needed for task-level autonomy, but populating them for an existing facility is still manual work. This paper presents an automated pipeline that extracts TOSM semantic objects from a single terrestrial laser scan and emits them as a VDA 5050-style JSON message. A deterministic stage segments the colored point cloud and fills the explicit model (pose, dimensions, color) with a provisional class from an open-vocabulary 3D classifier; a vision-language model (VLM) stage then verifies each class from multi-view object renders under forced tool-use and infers the implicit affordance model. We show that the obvious multi-view strategy — feeding all views into one prompt (early fusion) — *degrades* accuracy on sparse point renders, and propose per-view classification with confidence-weighted voting (late fusion) instead. On a 68.5M-point scan of a robot showroom (57 objects), late fusion cuts confidently-wrong labels from 24 (classifier-only) to 10, achieves the best lenient type accuracy (82%) among all evaluated configurations, recovers fine-grained classes such as keyboards, and provides a free reliability signal: unanimous votes across views are 94% correct. Full-scene verification costs \$2.6.

Index Terms—multi-view object recognition, vision-language model, semantic mapping, terrestrial laser scanning, TOSM, VDA 5050

I. INTRODUCTION

Task-level autonomy for indoor mobile robots requires more than an occupancy grid: the robot must know *what* objects occupy the space, *where* they are, and *how* they can be interacted with. The Triplet Ontological Semantic Model (TOSM) [1] organizes this knowledge into a *symbolic* model (class, name, identity), an *explicit* model (pose, size, color), and an *implicit* model (affordances such as whether an object is a reliable landmark or can be displaced). TOSM-style databases drive navigation frameworks and multi-robot fleets, and their content maps naturally onto the JSON message conventions of VDA 5050 [2], the emerging interface standard between AGV fleets and master control systems.

In practice, the semantic-object database for a new facility is authored by hand. Terrestrial laser scanners (TLS) such as the Leica BLK360 capture a dense, colored as-built model

in minutes, and scan-to-BIM research [3] recovers structural geometry automatically, but object-level semantics remain the gap: open-vocabulary 3D classifiers such as Uni3D [4] are biased toward industrial structural classes — in our experiments office chairs were labeled *stair* or *machine* and keyboards *ladder* — and no point-cloud classifier produces the affordance-level knowledge TOSM needs.

Vision-language models (VLMs) offer a complementary signal: rendered views of a segmented object carry texture and context that a point-cloud embedding discards. Since multi-view recognition classically outperforms single-view, we render each object from four azimuths and study how a VLM should consume them. The naive strategy — all views in one prompt (*early fusion*) — turns out to *hurt*: sparse oblique point renders invite over-confident misreadings that contaminate the joint judgment. Classifying each view independently and aggregating by confidence-weighted vote (*late fusion*), the multi-view analogue of MVCNN-style aggregation [5], restores and surpasses single-view reliability while exposing disagreement between views as an explicit uncertainty signal. The contributions are:

- an automated TLS-to-TOSM pipeline whose output is a VDA 5050-style `semanticObjects` message consumable by a fleet environment model;
- a schema-forced VLM verification protocol with per-view late fusion, plus prompt guardrails (size sanity, sparse-render caution, panel disambiguation) for point-render inputs;
- a six-configuration study on a real 68.5M-point BLK360 scan (57 objects) with a consistent manual audit, showing early fusion degrades accuracy while late fusion minimizes wrong labels (24→10) and yields 94%-precise unanimity gating.

II. RELATED WORK

Semantic environment models. TOSM [1] extends ontology-based robot knowledge bases [6] with the symbolic/explicit/implicit triplet; its records subsume the object fields of VDA 5050 [2] messages, which we adopt as the output container. This taxonomy also underpins a recent

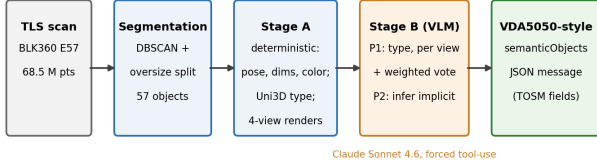


Fig. 1. Scan-to-semantics pipeline. Stage A is deterministic and fills the explicit TOSM model plus a provisional class; Stage B verifies the symbolic model per rendered view with confidence-weighted late fusion and infers the implicit model, all under forced tool-use.

knowledge-grounded 3D scene graph for LLM querying [7]; we address populating such records automatically from a TLS scan. **Scan-to-model.** As-built modeling from TLS has focused on structural elements [3]; object semantics are left to learned 3D segmentation [8] or open-vocabulary encoders such as Uni3D [4], which inherit training-corpus biases. **Multi-view recognition.** Aggregating rendered views is a classic route to 3D shape recognition (MVCNN [5]); we revisit the early-vs-late fusion question for VLM prompts over *sparse point* renders rather than mesh renders. **VLMs in robotics.** Recent systems use VLMs for open-set scene graphs [9] and language-driven maps [10]. We use a VLM in a narrower, auditable role — verifying one object at a time against measured geometry — keeping the deterministic stage authoritative for everything a VLM should not touch.

III. PIPELINE

A. Stage A: Deterministic Explicit Model

The input is a single registered BLK360 scan (E57, colored). After floor detection by RANSAC and wall removal, objects are segmented by DBSCAN [11] with a conservative re-clustering pass that splits clusters whose footprint exceeds 2.5m (30→57 objects on our scene). For each object Stage A computes the explicit TOSM model directly from the points: bounding-box center (`poseX/Y`, `properties.poseZ`), PCA-based yaw (`poseTheta`), yaw-aligned extents (`dimensions`), and a mean RGB color name. Uni3D provides a *provisional* symbolic class, and four azimuth renders (0/90/180/270°, 512×512, point size 8, red 3D bounding box) are exported. Render density matters: with the default small point size the same objects produced speckle-like views that misled both the VLM and the human auditors (Sec. IV-B). All Stage-A fields are measured, not generated; Stage B is never allowed to modify them.

B. Stage B: VLM Verification with Late Fusion

Stage B calls a commercial VLM (Claude Sonnet 4.6) with *forced tool-use*: the model must answer by calling a declared JSON tool, so every response is schema-valid by construction. Fig. 3 condenses the prompt design.

Prompt 1 (symbolic), once per view. The VLM sees ONE render, the measured dimensions, the provisional class, and an optional candidate vocabulary (*chair, TV, monitor, Mobile*

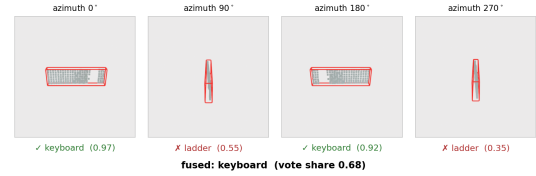


Fig. 2. Late fusion on one object (`keyboard_018`, provisional label *ladder*). Each of the four azimuth renders is classified independently; edge-on views vote for the wrong thin-strip class with low confidence and are outvoted by the high-confidence face-on views.

P1 system (condensed). You are a semantic-mapping verifier for an indoor mobile robot. You see a render of ONE segmented 3D object (sparse laser points, red 3D box), its classifier label `uni3d_type`, and optional candidates. **Decision rule:** (1) if `uni3d_type` is correct, KEEP; (2) if wrong, replace — (a) prefer a fitting candidate, (b) else the most accurate common-noun label. Never ‘unknown’. **Size sanity:** measured dimensions are a HARD constraint; a cluster exceeding a class’s envelope is merged objects — label the dominant structure or ‘clutter’. **Sparse-render caution:** never claim parts not clearly outlined by points. **Panel rule:** thin flat objects are keyboard (key grid) / display (uniform dark face) / shelf — never ladder. **Chair gate:** requires visible seat pan 0.3–0.6m high plus backrest. If uncertain, keep a size-compatible label. Respond ONLY via the `report_symbolic` tool.

P2 system (condensed). Infer implicit properties of ONE object from its views and confirmed type: `isKeyObject` (fixed, distinctive landmark), `isMovable`, `isOpen/canBeOpen` (doors only, else null). Respond ONLY via the `report_implicit` tool.

Fig. 3. Condensed Stage-B prompt design. Forced tool-use makes every response schema-valid JSON; measured geometry is read-only context.

Robot, desk). The decision rule is three-way: keep a correct label; if wrong, prefer a fitting candidate; else emit an open-vocabulary label. Guardrails address point-render failure modes: the measured dimensions are a hard constraint (a 1.6m box is never a *chair*); clusters larger than any single-instance envelope are treated as merged objects; thin flat objects are disambiguated among keyboard / display / shelf by their face pattern; sparse-render caution forbids claiming parts that are not clearly outlined by points; and when no label is clearly supported, the rule prefers keeping a size-compatible provisional label or falling back to *clutter*.

Late fusion. The four per-view reports are aggregated by confidence-weighted voting: each vote adds its confidence to its label’s score, the top-scoring label wins, and the final confidence is the winner’s share of the total. The per-view votes are stored in the record (`properties.viewVotes`), so a downstream consumer can distinguish a 4/4 unanimous label from a 2/4 split — an uncertainty signal that costs nothing extra. Fig. 2 shows the mechanism on a real object: the face-on views identify a keyboard with high confidence, the edge-on views misread the thin silhouette as *ladder*, and the weighted vote resolves the split correctly — the same views that, concatenated into one early-fusion prompt, drag the joint judgment toward the wrong label.

Prompt 2 (implicit), once per object. Given the fused type and all views, the VLM reports the TOSM implicit model: `isKeyObject`, `isMovable`, and (doors only) `isOpen/canBeOpen`.

TABLE I
TOSM $\rightarrow$ VDA 5050-STYLE JSON FIELD MAPPING

TOSM model	element	JSON field
symbolic	class	type
	name / identity	name, id
explicit	position (x, y)	poseX, poseY
	height z	properties.poseZ
	orientation	poseTheta
	size	dimensions{l, w, h}
	color	color
implicit	landmark / movability	properties.isKeyObject,
	door state	isMovable, isOpen, canBeOpen
–	VLM audit trail	properties.viewVotes,
		symbolicReason, confidence

C. TOSM to VDA 5050-Style Message

The completed records are serialized as one `semanticObjects` JSON message with a VDA 5050-style header. Table I shows the field mapping; TOSM fields with no standard slot travel in the extensible `properties` map, so the message remains readable by consumers that only know the base schema.

IV. EXPERIMENTS

A. Setup and Audit Protocol

We evaluate on a single BLK360 scan of an indoor robot showroom (68.5M colored points, 57 segmented objects) with Claude Sonnet 4.6. Six configurations are compared (Table II): the classifier alone; single-view and four-view early-fusion prompts, each with the initial sparse renders/base prompt and with dense renders/guardrail prompt; and four-view late fusion. To grade them consistently, two of the authors audited all 57 objects against the *dense* four-view renders and the measured dimensions, rating every configuration’s final label *correct*, *plausible* (consistent with the views but not verifiable from renders alone), or *wrong*. Metered cost: late fusion \$2.63 per scene (285 calls, \$0.046/object); early fusion \$1.27; single view \$0.87.

B. Multi-View Study

Table II and Fig. 5(b) summarize the audit. Three findings stand out.

(1) *Early fusion hurts*. Feeding four sparse views into one prompt dropped strict accuracy from 60% (single view) to 49%: extra oblique speckle views encouraged over-committed readings, e.g. two electrical cabinets confidently relabeled *chair*. Denser renders and the prompt guardrails shrank but did not close the gap (53% vs. 58%).

(2) *Late fusion makes multi-view the most reliable configuration*. Judging each view independently and voting yields the fewest confidently-wrong labels of all configurations (10 of 57, vs. 11–19 for the VLM alternatives and 24 for the classifier alone) and the highest lenient accuracy (82%). A contaminated view can no longer dominate: it is outvoted. The cost is a larger *plausible* bucket — split votes often land on the generic *clutter* — which trades decisiveness for safety, the right trade for an autonomously written robot database.

TABLE II
MANUAL AUDIT OF FINAL TYPE LABELS (57 OBJECTS). C/P/W = CORRECT / PLAUSIBLE / WRONG; STRICT = C, LENIENT = C+P.

Configuration	C	P	W	strict	lenient
Uni3D only (no VLM)	28	5	24	49%	58%
1 view, base prompt, sparse	34	12	11	60%	81%
4 views early fus., sparse	28	10	19	49%	67%
1 view, guardrails, dense	33	13	11	58%	81%
4 views early fus., dense	30	13	14	53%	75%
4 views late fus., dense	32	15	10	56%	82%

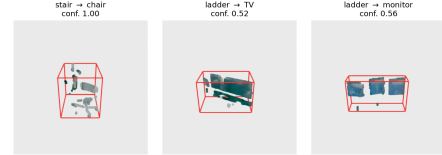


Fig. 4. Representative late-fusion corrections (one of four dense views; provisional label $\rightarrow$ fused label, vote-share confidence). The 3D classifier’s industrial bias (*stair*, *ladder*) is corrected to furniture and displays; the keyboard recovery is detailed in Fig. 2.

(3) *Vote unanimity is a free precision signal*. 17 of 57 objects received identical labels from all four views; 16 of those (94%) were audited correct, versus 40% among split-vote objects. Gating on the stored `viewVotes` thus gives a downstream consumer a calibrated route-to-human-review policy that per-call confidence alone did not provide.

Render density mattered twice over: dense renders let the VLM recover four keyboards that sparse renders had disguised as *monitor* or *ladder* strips (Fig. 2), and they corrected the human audit itself — labels initially graded correct from sparse renders turned out wrong once the key grids became visible. Run-to-run sampling variance (± 2 –3 objects) bounds how finely the configurations can be ranked.

C. Implicit Model and Completed Records

On the late-fusion output, Prompt 2 marked 6 of 57 objects as key objects (control panels, a large machine, mobile robots, a wall display) and 50 as movable. Fig. 6 shows two completed TOSM records as stored in the output message, including the audit trail (per-view votes, fused confidence, one-sentence reason). Mean fused confidence was 0.71, and the 17 unanimous objects carry confidence 1.0 by construction, aligning the stored confidence with the observed 94% unanimity precision.

D. Discussion and Limitations

The division of labor is the main design point: everything measurable stays deterministic and VLM-immutable, so hallucination can corrupt neither poses nor dimensions, and the VLM contributes exactly the judgments geometry cannot make — naming and affordances — in a form (per-view votes, forced-tool JSON, stored reasons) that is auditable after the fact. All 741 tool calls across the five VLM configurations returned schema-valid JSON. Remaining errors concentrate where the render itself is ambiguous: texture-poor slabs, a merged chair-plus-robot cluster from DBSCAN, and sparse

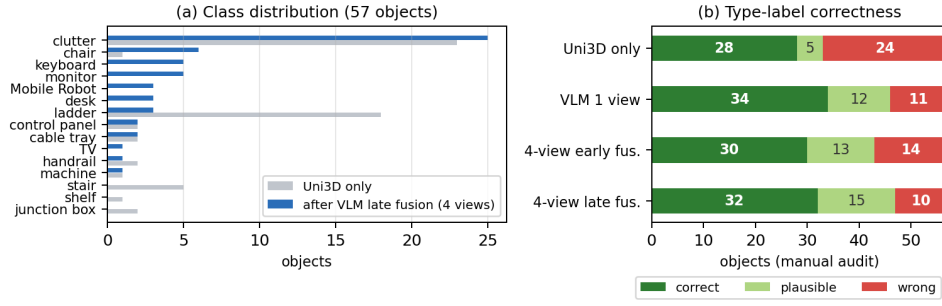


Fig. 5. Effect of VLM verification on the 57-object scene. (a) Class distribution, classifier-only vs. after late fusion: implausible *ladder/stair* populations are redistributed to displays, chairs, and keyboards. (b) Manual audit: early fusion degrades single-view accuracy, while late fusion yields the fewest wrong labels (10) and the best lenient accuracy (82%).

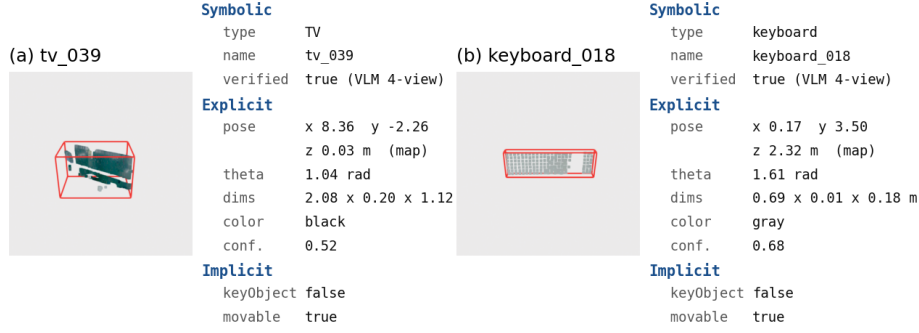


Fig. 6. Completed TOSM semantic-object records (dense render + symbolic / explicit / implicit fields) as serialized into the VDA 5050-style message. (a) A wall display corrected from *ladder*. (b) A keyboard recovered from *ladder*, an open-vocabulary class outside the candidate list.

strips. These point to upstream fixes — finer segmentation, mesh renders — not to the fusion protocol. The audit was performed by the authors from the same dense renders the VLM saw; an on-site inventory would tighten the numbers. Absolute strict accuracy (~56–60%) reflects a deliberately hard test bed: a cluttered showroom scanned once, without manual cleanup.

V. CONCLUSION

We presented an automated pipeline that turns a single terrestrial laser scan into TOSM semantic-object records delivered as a VDA 5050-style JSON message, using a VLM in a tightly scoped, schema-forced role. A six-configuration study showed that the intuitive multi-view prompt — all views at once — degrades verification on sparse point renders, whereas per-view classification with confidence-weighted late fusion makes multi-view the most reliable configuration: wrong labels drop from 24 (classifier-only) to 10, lenient accuracy reaches 82%, and vote unanimity provides a 94%-precise gating signal at \$2.6 per scene. Future work includes mesh-based renders, finer instance segmentation, and coupling the records to place modeling and a live digital twin of the same facility.

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